Accelerated Semiconductor R&D: AI Surrogate Modeling

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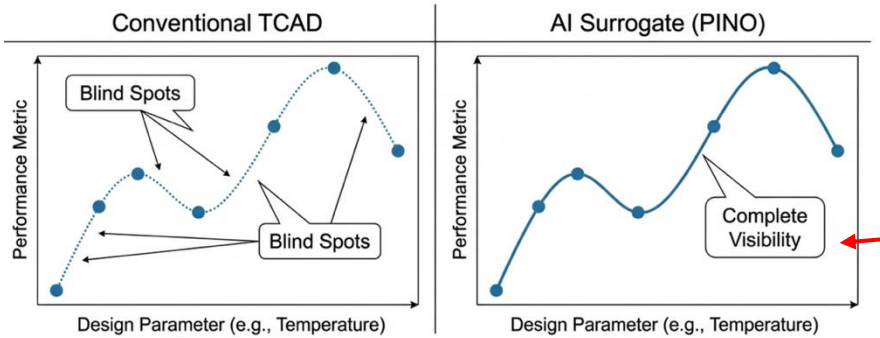
ELECTRICAL [+] COMPUTER ENGINEERING

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Physics-Informed AI to break through R&D Bottleneck

- **The Cost of Precision:** Conventional TCAD simulations for advanced Memory (NAND/DRAM) are computationally prohibitive (hours/days per run).
  - **Optimization "Blind Spots":** Sparse data points lead to missed opportunities in design space exploration between simulated nodes.
  - **Static Methodology:** Altering a single geometry parameter (e.g., capacitor aspect ratio) requires a total mesh rebuild and re-simulation.
- **Beyond "Black Box" AI:** Implementing PINO (Physics-Informed Neural Operators)
  - Introducing Physics Loss (Poisson, Arrhenius, etc.) within the model architecture.



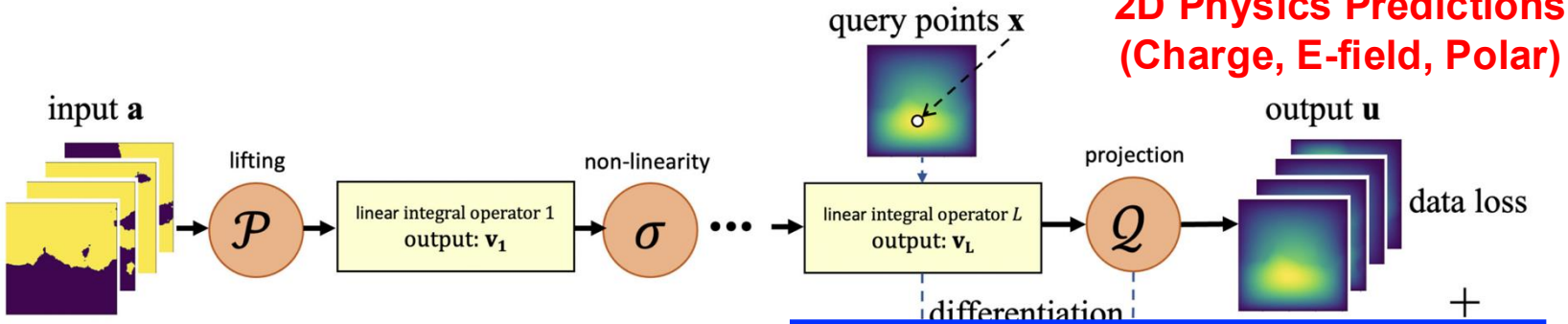
$$\mathcal{L}_{total} = \underbrace{\|\hat{u} - u_{target}\|_{L_1}}_{\text{Data Accuracy}} + \lambda \cdot \underbrace{\|\nabla\hat{\phi} - \nabla\phi_{target}\|_{L_1}}_{\text{Physical Consistency}}$$

(Matching sparse TCAD ground truth) (Penalizing PDE violations)

PINO Architecture

- Training Data: Fully compatible with conventional TCAD.

Output:
2D Physics Predictions
(Charge, E-field, Polar)



Input:
Device Geometry
Experimental Condition
Material Properties

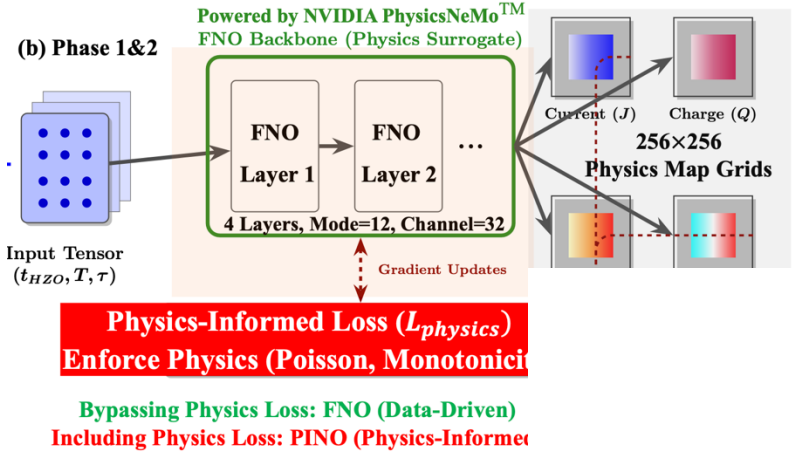
Physics-informed Loss function

e.g., Poisson Equation (Charge-Field Derivative Relationship)

Zongyi Li et al, Physics-Informed Neural Operator for Learning Partial Differential Equations

PINO Architecture

- FNO: Learning solution operator of the governing PDEs
  - Analogy: Vision model for scientific field simulation.
- PINO: FNO + Physics Loss



B. Physics-Informed Loss Function

A fundamental distinction between the standard, purely data-driven FNO and physics-informed PINO is the implementation of a dual-loss formulation during the training phase. The overall objective function is defined as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \mathcal{L}_{physics} \quad (1)$$

The data loss term (L_{data}) computes the L2 norm error between the neural network's predicted 2D physical fields or I_D-V_G curves and the TCAD-generated ground truth. To transition the model from a black-box approximator to a PINO, the physics loss ($L_{physics}$) is introduced as a regularizer. This term enforces physical consistency, which is necessary to constrain the hypothesis space in the absence of dense training data. We employed two primary $L_{physics}$ components: Poisson residual and monotonicity constraints.

1. **Poisson Residual**
   This term penalizes deviations from the governing electrostatics. It ensures that the predicted electrostatic potential (ϕ), charge density (ρ), and ferroelectric polarization (P) follow Poisson's equation:

$$\mathcal{L}_{Poisson} = \nabla \cdot (\epsilon \nabla \phi) + (\rho - \nabla \cdot \mathrm{P}) \quad (2)$$

Gyujun Jeong et al, Physics-informed AI Surrogate Modeling for Accelerated Retention Prediction in Ferroelectric Vertical NAND, ICMC 2026, submitted.

Governing PDE: Poisson Equation

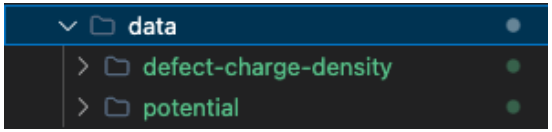
- Poisson Equation: $\nabla \cdot (\varepsilon \nabla \Phi) + \rho = 0$
  - Required Data: Potential, Permittivity, Charge Density
  - Φ: Potential Map
  - ε: Permittivity (e.g., varies across the Si, SiO2, Si3N4 layers)
    - Will use ε_{Si3N4} = 7.5 for the brevity of demo
  - ρ: Defect charge density (= Charge Migration)

- 2D Spatial (R, x) PDE: $\epsilon \left(\frac{\partial^2 \Phi}{\partial R^2} + \frac{\partial^2 \Phi}{\partial x^2} \right) + \rho = 0$

```
# POISSON EQUATION: \epsilon * (\nabla^2 \phi) + \rho = 0
poisson_pde = self.epsilon * (d2phidr2 + d2phidx2) + rho
```

```
poisson_node = Node(
    inputs=inputs,
    outputs=["poisson"],
    evaluate=Poisson(gradient_method=cfg.custom.gradient_method, epsilon_rel=7.5),
    name="Poisson Node",
)
```

PDE Definition



Training Data Structure

```
#HEADER
Time (s),0
#DATA
Ryz (m),X (m),Defect Charge Density (C \cdot cm^{-3})
4e-9,-7.75e-8,-1.2966768510302977
4e-9,-7.527777777777778e-8,-1.2909394752096262
4e-9,-7.305555555555556e-8,-1.2734033598679972
```

Potential Map

```
#HEADER
Time (s),0
#DATA
Ryz (m),X (m),Potential (V)
4e-9,-7.75e-8,-1.32635027854983
4e-9,-7.527777777777778e-8,-1.3208271582084683
4e-9,-7.305555555555556e-8,-1.2887939188974897
```

Charge Density Map

Example: 3D NAND Charge Migration Modeling

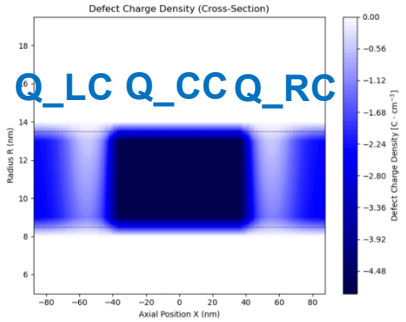
- **Retention loss due to charge migration**
  - **Input Variables (according to IRPS 2019, Data from Applied Materials Ginestra™):**
    - **1. Tunnel layer Thickness (t_{TL})**
      - t_TL = 2, 2.5, 3, 3.5, 4nm to
    - **2. PGM voltage (V_{PGM})**
      - V_PGM = 10, 15, 20V
    - **3. Cell Spacing (L_{SP})**
      - L_sp = 20, 25, 30nm

→ **PINO Map**

Output: Retention Charge Map $Q = \int \rho_{def}(r,t)dr$

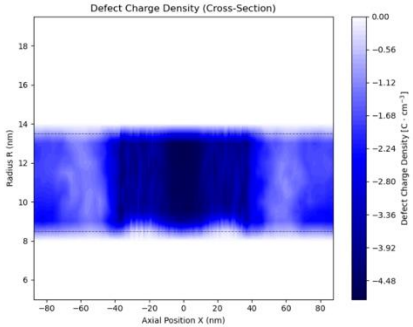
→ $Q_{Migrate} = \Delta(Q_{LC}+Q_{RC})[t_{ret} - t_0]$

$w.r.t.\{t_{TL}, V_{PGM}, L_{SP}\}$

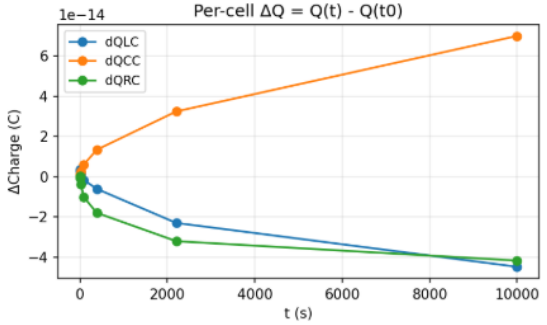


t=0

Retention Loss



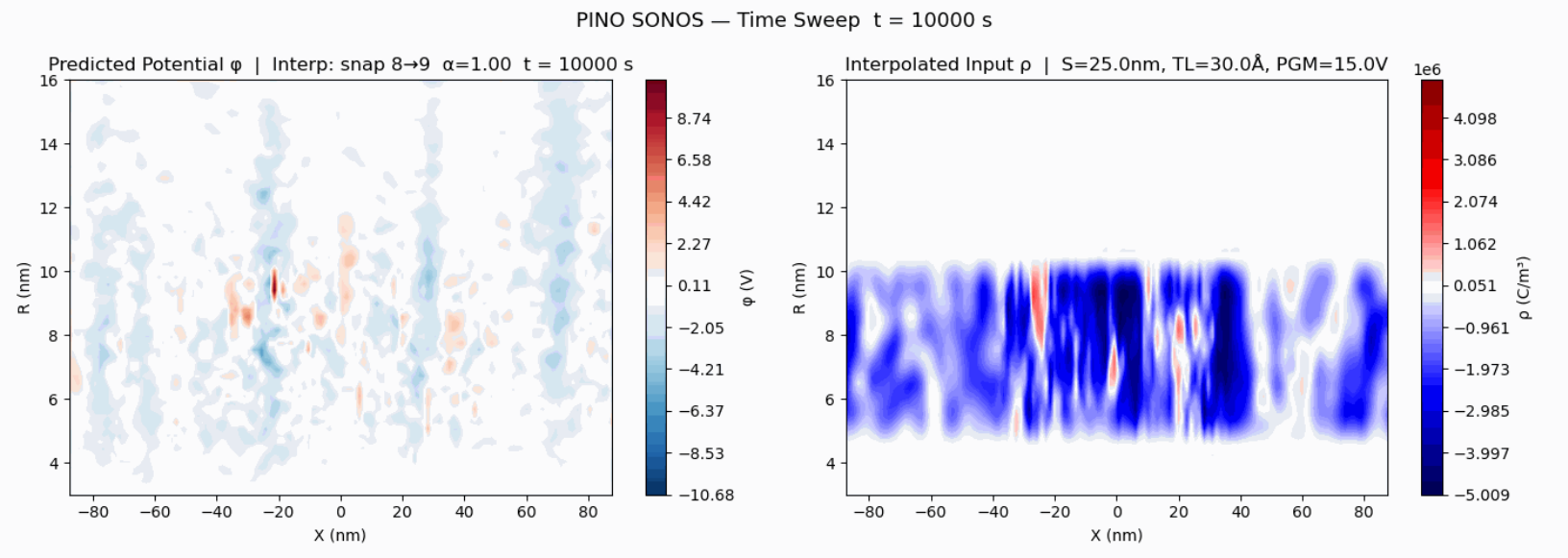
t=1e4s (=t_ret)



Calculated Retention Loss

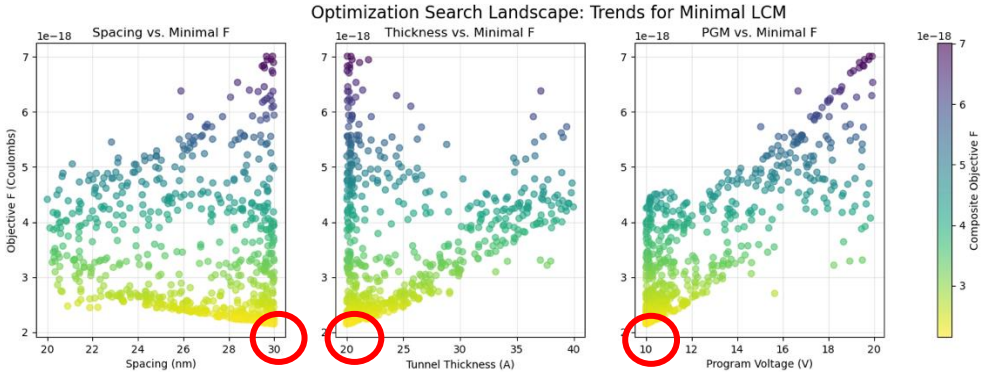
3D NAND Charge Migration Modeling

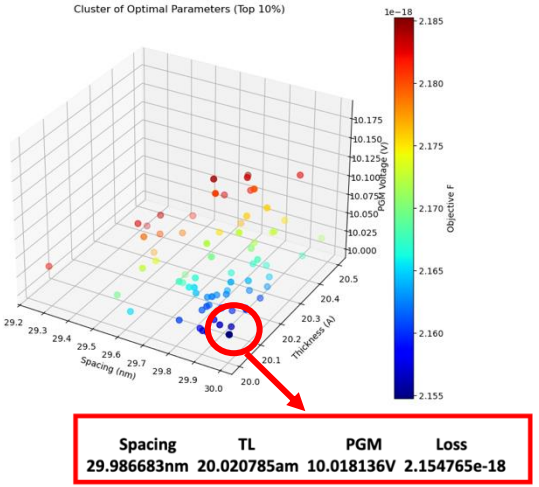
- **Advantage 1) Accelerated Simulation**
  - Interpolation visualization available
  - Time advantage: 3 days+ for discrete sweeps vs. inference less than milliseconds



3D NAND Charge Migration Modeling

- **Advantage 2) Multi-parameter Optimization**
  - Can observe physically reliable result
  - Can expand this framework for non-linear variables optimization





Variables with physical interpretation (IRPS 2019):

1. Wide Cell Spacing → Prevents LCM
2. Thick Tunnel Layer Thickness → LCM ratio increases
3. High PGM Voltage → LCM-dominant